

# Model Hybrid Connectivity and BGP Failure

90 minutes · Mapped criteria A4, K5

## Purpose and work product

Evaluate VPN versus Interconnect, inspect mock BGP routes and isolate a hybrid routing failure.

**Submit:** connectivity decision, route analysis and corrected BGP configuration

## Safety, cost and privacy

Use only the assigned training project. Confirm the active account, project and region before every mutation. Never place credentials, payment details, access tokens or private keys in screenshots or submitted files. Stop billable resources and verify the final inventory.

## Files in this folder

| File               | Purpose                                                  |
|--------------------|----------------------------------------------------------|
| bgp_routes.csv     | Mock data, script or configuration used by this activity |
| router_config.yaml | Mock data, script or configuration used by this activity |

## Detailed procedure

### Step 1

Read the scenario: an on-premises 172.16.0.0/16 network must reach ace-vpc with predictable failover.

### Step 2

Complete a decision table comparing HA VPN, Partner Interconnect and Dedicated Interconnect for throughput, SLA, lead time and cost.

### Step 3

Inspect router\_config.yaml and confirm local ASN 64514 differs from peer ASN 65010.

### Step 4

Open bgp\_routes.csv and identify the most specific route for 10.20.10.0/24.

### Step 5

Simulate a missing 10.20.10.0/24 advertisement and explain the traffic consequence.

### Step 6

Use a trainer sandbox or mock gcloud output to inspect VPN tunnel status and Cloud Router BGP peer status.

### Step 7

Trace one packet from 172.16.0.10 to 10.20.10.20 through peer gateway, tunnel, Cloud Router, VPC route and firewall.

### Step 8

Correct the advertisement or route priority in a copy of the YAML/CSV.

### Step 9

Run a configuration comparison and save the before/after result.

### Step 10

Prepare a short hybrid-network incident note including the failed hop, evidence and escalation owner.

## Verification record

Record expected and observed output. A successful command without the correct project, identity, scope or evidence is not an accepted result.

| Check              | Expected evidence                                      |
|--------------------|--------------------------------------------------------|
| Identity and scope | Active account, project ID, region or zone             |
| Configuration      | Resource description or policy output                  |
| Runtime result     | Test output with UTC timestamp                         |
| Failure test       | One controlled negative test and diagnosis             |
| Cleanup            | Final inventory and stopped/deleted billable resources |

## Troubleshooting sequence

Check the active identity and project first, then API enablement and IAM, then resource state, networking and quotas. Preserve the failed output before making a correction. If the issue remains, escalate with the exact command, timestamp, expected result, observed result and evidence files.

## Cleanup

Delete or stop resources as instructed. Re-open Billing and the relevant service inventory, then record the remaining resource names. Budget alerts do not stop charges.